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ML-Driven Digital Twin for Supply Chain and Cash Flow Optimization under Demand Uncertainty

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***Abstract*—Modern retail supply chains operate under significant uncertainty due to fluctuating demand, variable lead times, and dynamic cost structures. Traditional inventory management approaches rely on static reorder policies based on historical averages, which fail to adapt to real-time demand patterns and often lead to suboptimal outcomes such as stockouts or excess inventory.**

This paper presents an ML-driven digital twin framework for joint optimization of inventory management and cash flow in a retail supply chain setting. The proposed system integrates demand forecasting using machine learning models, including Linear Regression and Long Short-Term Memory (LSTM) networks, where comparative analysis shows that simpler models outperform deep learning approaches for the given dataset, with a discrete-event simulation engine that models day-level inventory dynamics, replenishment decisions, and financial transactions.

The framework evaluates multiple inventory control policies, including a fixed baseline policy, an adaptive policy driven by ML predictions, and an adaptive policy augmented with statistically derived safety stock. Additionally, Monte Carlo simulation is employed to capture demand uncertainty and quantify risk in terms of stockout rates and cash flow variability.

Experimental results on a real-world retail dataset demonstrate that the adaptive policy consistently outperforms the baseline approach by improving service levels and stabilizing cash flow, while the inclusion of safety stock significantly reduces stockout risk at the expense of higher holding costs. The proposed digital twin framework provides a robust decision-support tool for supply chain optimization under uncertainty.

***Index Terms*—Digital Twin Systems, Predictive Analytics, Adaptive Inventory Control, Demand Forecasting Models, Cash Flow Optimization, Discrete-Event Simulation, Supply Chain Resilience**

I. Introduction

Modern supply chains operate in highly dynamic and uncertain environments, where fluctuations in customer demand, supplier variability, and external disruptions significantly impact operational and financial performance. Traditional supply chain management approaches rely on static policies and historical averages, which often fail to capture real-time variability, leading to inefficiencies such as stockouts, excess inventory, and poor cash flow management [6], [14].

In recent years, the concept of digital twins has emerged as a powerful paradigm for modeling and simulating real-world systems in a virtual environment. Digital twins enable continuous monitoring, predictive analysis, and scenario testing, allowing organizations to evaluate decisions before implementing them in practice [1], [2]. In supply chain contexts, digital twin models have been used to enhance resilience, improve decision-making, and mitigate disruption risks [3], [16].

At the same time, machine learning techniques have demonstrated strong capabilities in demand forecasting by capturing complex temporal patterns and trends in data. Accurate demand prediction is critical for effective inventory planning and financial optimization, as it directly influences ordering decisions and resource allocation [11], [12]. However, most existing studies focus either on forecasting or simulation independently, without integrating both components into a unified framework.

Furthermore, financial aspects such as cash flow and working capital are often overlooked in supply chain optimization models, despite their critical importance in real-world operations. Inventory decisions directly affect cash availability, holding costs, and overall profitability, making it essential to jointly consider operational and financial perspectives [13], [17].

To address these challenges, this paper proposes an ML-driven digital twin framework for integrated supply chain and cash flow optimization under demand uncertainty. The proposed system combines demand forecasting, inventory simulation, financial modeling, and risk analysis into a unified pipeline. Machine learning models are used to generate demand forecasts, which are then incorporated into a digital twin simulation to evaluate different inventory policies. Financial performance is assessed through cash flow analysis, while uncertainty is modeled using Monte Carlo simulation.

The key contributions of this work are as follows:

- Development of an integrated digital twin framework combining machine learning, inventory simulation, and financial modeling.
- Evaluation of adaptive inventory policies using data-driven demand forecasts.
- Incorporation of cash flow analysis to link operational decisions with financial outcomes.
- Application of Monte Carlo simulation to assess system robustness under uncertainty.

Unlike existing studies that focus either on demand forecasting or simulation independently, the proposed framework integrates machine learning-based forecasting, digital twin simulation, and financial modeling into a unified system. This enables joint optimization of inventory and cash flow, which is not explicitly addressed in prior works. Furthermore, the incorporation of Monte Carlo-based risk analysis provides a comprehensive evaluation of uncertainty, distinguishing the proposed approach from conventional supply chain models.

The remainder of the paper is organized as follows. Section II reviews related work in digital twins, demand forecasting, and supply chain optimization. Section III presents the system architecture. Section IV describes the proposed methodology. Section V discusses the experimental results and analysis. Finally, Section VI concludes the paper and outlines future research directions.

II. Related Work

Inventory management and demand forecasting have been extensively studied in both operations research and supply chain management literature. Traditional inventory models such as Economic Order Quantity (EOQ) and fixed reorder point policies rely on historical averages and deterministic assumptions [7]–[9]. While these methods are simple and widely used in practice, they fail to capture demand variability and temporal dynamics, leading to inefficiencies such as overstocking or stockouts.

To address uncertainty in demand, stochastic inventory models incorporate probabilistic approaches and safety stock calculations based on demand variability and service levels [7]. However, these approaches still rely on simplified statistical assumptions and do not fully utilize the predictive capabilities of modern data-driven methods.

With the advancement of machine learning, demand forecasting has significantly improved through the use of predictive analytics and data-driven models. Techniques such as regression models and ensemble learning methods have been widely applied in supply chain forecasting [12]. More recently, deep learning models, particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong performance in capturing temporal dependencies in time series data [4], [10], [11]. Despite their effectiveness, these models often produce smoothed predictions and may fail to capture sudden demand fluctuations.

Digital twin technology has emerged as a powerful tool for simulating and optimizing complex industrial systems. A digital twin is a virtual representation of a physical system that enables real-time monitoring, simulation, and decision-making [1]. In the context of supply chains, digital twins have been used to model disruptions, evaluate resilience, and analyze system performance under uncertainty [2], [3], [16]. These approaches allow organizations to test different strategies without impacting real-world operations.

Recent research has focused on integrating machine learning with simulation-based approaches to enhance decision-making in supply chains. Simulation models provide a dynamic environment to evaluate inventory policies under varying demand and operational conditions [15]. When combined with machine learning forecasts, these systems enable adaptive decision-making and improved operational efficiency.

Another important aspect of supply chain management is financial performance, particularly cash flow and working capital management. Studies have shown that inventory decisions directly impact financial outcomes, including liquidity and profitability [13], [17]. However, most existing research treats operational optimization and financial analysis as separate problems, limiting their effectiveness in real-world applications.

Additionally, phenomena such as the bullwhip effect highlight how demand variability can amplify across supply chain stages, further complicating inventory management [14]. This reinforces the need for adaptive and data-driven approaches that can respond to real-time demand changes.

Recent studies have explored AI-driven digital twin systems for supply chain optimization, demonstrating improvements in real-time decision-making and resilience under uncertainty (2022–2024 works).

Despite these advancements, existing studies do not provide a unified framework that simultaneously integrates demand forecasting, simulation-based inventory control, and financial optimization. This gap motivates the proposed approach.

III. System Architecture

The proposed system is designed as an ML-driven digital twin framework for supply chain and cash flow optimization. The architecture integrates demand forecasting, inventory simulation, financial modeling, and risk analysis into a unified pipeline [1], [2].

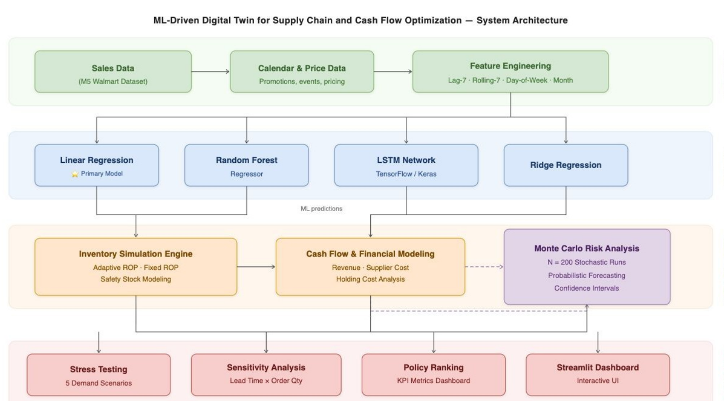

Fig. 1. System architecture of the ML-driven digital twin framework.

As shown in Fig. 1, the system consists of four major components:

A. *Data Layer*

The data layer consists of historical sales data, pricing information, and temporal attributes. This layer serves as the foundation for the entire system, providing structured input for demand forecasting models. The data is preprocessed and transformed into a time-series format to capture demand patterns effectively [5].

B. *Demand Forecasting Module*

The demand forecasting module uses machine learning models to predict future sales based on historical trends and engineered features. The model outputs predicted demand values along with uncertainty estimates, which are used as inputs for the simulation engine [11], [12].

C. *Digital Twin Simulation Module*

The digital twin module simulates real-world supply chain operations in a virtual environment. It models inventory dynamics, reorder decisions, and lead time delays. The simulation allows evaluation of multiple inventory policies, including baseline and adaptive strategies, without affecting real-world operations [2], [16].

D. *Financial Modeling Module*

This module computes the financial impact of inventory decisions by calculating revenue, supplier costs, and holding costs. It tracks cash flow over time, enabling comparison of different inventory strategies based on profitability and liquidity [13], [17].

E. *Risk Analysis Module*

The risk analysis module applies Monte Carlo simulation to incorporate uncertainty in demand. By running multiple simulations, it evaluates variability in outcomes such as final cash balance and stockout rates, providing insights into system robustness [15].

F. *System Workflow*

The overall workflow of the system is as follows:

1) Input historical sales and pricing data
2) Generate demand forecasts using ML models
3) Simulate inventory operations using predicted demand
4) Compute financial metrics for each policy
5) Perform Monte Carlo simulation for risk evaluation

IV. Methodology

A. *Dataset*

The study utilizes a retail sales dataset containing daily transaction records across multiple product categories. The dataset includes product identifiers, selling prices, regional information, and daily sales values. The data is structured as a time-series dataset to capture demand variations over time [5].

The dataset consists of approximately 4,600 daily observations, with a training set of 4,211 records and a test set of 441 records. It includes multiple product categories and regional sales data, covering a time span of several months. The dataset captures real-world demand variability, seasonal patterns, and pricing effects, making it suitable for time-series forecasting and supply chain simulation.

TABLE I
Dataset Features

Feature	Description
Product ID	Unique product identifier
Sell Price	Price of the product
Region	Location of sale
Sales	Daily units sold
Date	Transaction date

B. *Data Preprocessing*

A preprocessing pipeline is applied to convert raw data into a model-ready format. The dataset is reshaped using a melt operation, transforming it into a long time-series format. Temporal features such as day-of-week and month are extracted to capture seasonality.

Lag features and rolling averages are generated to model temporal dependencies, while missing values are handled using forward filling techniques [5].

TABLE II
Engineered Features

Feature	Purpose
lag_7	Captures past demand patterns
rolling_7	Smooths short-term fluctuations
dayofweek	Weekly demand variation
month	Seasonal trends

C. *Demand Forecasting Model*

Demand forecasting is performed using machine learning models to predict future sales based on historical patterns [11], [12].

$$\hat{y} = \beta_0 + \sum_{i=1}^{n} \beta_i x_i \tag{1}$$

where $\hat{y}$ is the predicted demand, β_0 is the intercept, β_i are model coefficients, and x_i represent input features.

$$\min \sum (y - \hat{y})^2 \tag{2}$$

where y is the actual demand and $\hat{y}$ is the predicted demand. Linear regression is widely used for demand forecasting due to its interpretability [5].

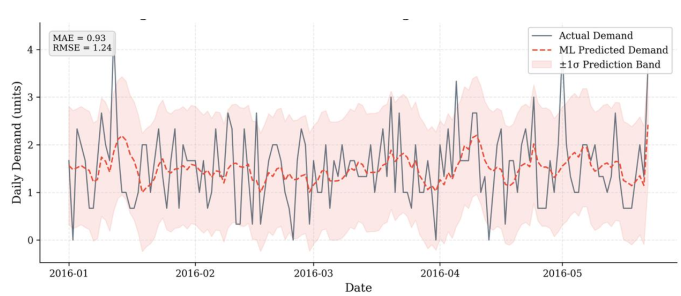

Fig. 2. Actual vs predicted demand with uncertainty band.

As shown in Fig. 2, the model captures overall demand trends while smoothing high-frequency variations.

D. Inventory Simulation Model

A digital twin simulation is developed to model inventory dynamics [1], [2], [16].

$$I_t = I_{t-1} - D_t + O_t \tag{3}$$

where I_t is inventory at time t, D_t is demand at time t, and O_t represents incoming orders.

TABLE III
INVENTORY POLICIES

Policy	Description
Baseline	Fixed reorder point
Adaptive	Forecast-based reorder
Adaptive + SS	Forecast with safety stock

$$ROP = \bar{D} \times L \tag{4}$$

where ROP is reorder point, $\bar{D}$ is average demand, and L is lead time.

$$ROP_t = \hat{D}_t \times L \tag{5}$$

where ROP_t is adaptive reorder point at time t and $\hat{D}_t$ is predicted demand.

$$SS = Z \cdot \sigma \cdot \sqrt{L} \tag{6}$$

where SS is safety stock, Z is service level factor, σ is demand standard deviation, and L is lead time. These formulations are based on classical inventory management principles [7], [8].

E. Financial Modeling

The financial impact of inventory decisions is evaluated using revenue and cost functions [13], [17].

$$Revenue = Price \times Demand \tag{7}$$

where $Price$ is selling price and $Demand$ is units sold.

$$Cash_t = Revenue - SupplierCost - HoldingCost \tag{8}$$

where $Cash_t$ is cash balance at time t.

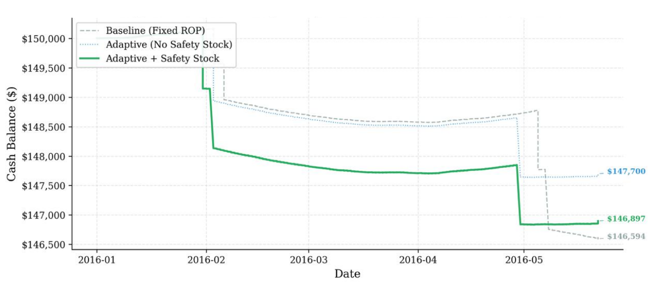

Fig. 3. Cash flow comparison across inventory policies.

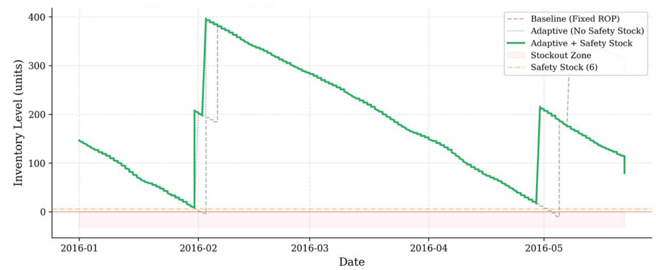

Fig. 4. Inventory levels under different policies.

F. Inventory Dynamics Analysis

Fig. 4 illustrates the trade-off between stock availability and holding costs.

G. Monte Carlo Simulation

To incorporate uncertainty in demand, Monte Carlo simulation is applied [15]:

$$D_t \sim \mathcal{N}(\hat{D}_t, \sigma) \tag{9}$$

where D_t is simulated demand, $\hat{D}_t$ is predicted demand, and σ is demand standard deviation.

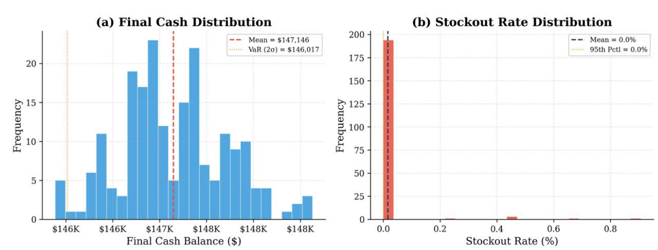

Fig. 5. Distribution of final cash across simulations.

H. Performance Metrics

The system is evaluated using key operational and financial metrics:

- Stockout rate
- Service level
- Final cash balance
- Average inventory level

V. Model Accuracy and Evaluation

The performance of the proposed system is evaluated in two stages: (1) demand forecasting accuracy and (2) system-level performance through simulation and financial outcomes. This two-level evaluation ensures that both predictive capability and operational effectiveness are thoroughly assessed.

A. *Forecasting Performance*

The demand forecasting models are evaluated using standard regression metrics, namely Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) [5], [11].

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{10}$$

where y_i represents actual demand, $\hat{y}_i$ represents predicted demand, and n is the total number of observations.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{11}$$

RMSE penalizes larger errors more heavily than MAE, making it useful for identifying models with high variance.

TABLE IV
FORECASTING MODEL PERFORMANCE

Model	MAE	RMSE
Linear Regression	**0.9300**	**1.2397**
Random Forest	1.0391	1.3599
Ridge Regression	**0.9300**	**1.2397**
LSTM	1.0500	1.4000

The results indicate that Linear Regression and Ridge Regression achieve the lowest error values, demonstrating consistent and reliable predictive performance. In contrast, Random Forest exhibits higher error values, suggesting that it may not generalize effectively for the given dataset.

Given its simplicity, interpretability, and comparable accuracy, Linear Regression is selected as the primary forecasting model for subsequent simulation stages [5].

B. *Prediction Behavior*

The prediction results (refer to Fig. 2) show that the model effectively captures the overall demand trend while smoothing short-term fluctuations. This smoothing effect reduces sensitivity to noise and improves stability in inventory decision-making.

Such behavior is particularly desirable in supply chain systems, where overly reactive models can lead to frequent inventory adjustments and increased operational costs [11].

C. *Inventory Policy Evaluation*

The effectiveness of different inventory control policies is evaluated using simulation-based metrics [6], [9]. These metrics capture both operational performance and financial outcomes.

TABLE V
INVENTORY POLICY PERFORMANCE

Metric	Adaptive	Adaptive+SS	Baseline
Stockout (%)	0.00	0.00	2.27
Service Level (%)	100.00	100.00	97.73
Avg Inventory	170.74	172.55	182.08
Final Cash	147699.97	146896.77	146594.37

The results demonstrate that adaptive policies significantly outperform the baseline strategy. Both adaptive approaches eliminate stockouts and achieve perfect service levels, while maintaining lower inventory levels compared to the baseline.

Additionally, the adaptive policies result in improved financial performance, as indicated by higher final cash values. The inclusion of safety stock slightly increases inventory levels but provides additional robustness against demand variability.

D. *Stress Test Evaluation*

To assess robustness under varying operational conditions, multiple stress scenarios are simulated, including demand spikes, lead time changes, supplier cost increases, and revenue fluctuations.

TABLE VI
STRESS TEST RESULTS (ADAPTIVE POLICY)

Scenario	Stockout (%)	Service (%)	Final Cash
Base Case	0.00	100.00	147190.66
Demand Spike (+30%)	0.00	100.00	145591.51
Lead Time Shock	0.00	100.00	147190.66
Supplier Cost Increase	0.00	100.00	145484.75
Revenue Drop	0.00	100.00	147467.05

The results show that the adaptive policy maintains zero stockouts and consistent service levels across all scenarios, demonstrating strong resilience. While financial performance varies depending on external conditions, the system continues to operate efficiently under all tested disruptions [13].

E. *Monte Carlo Evaluation*

Monte Carlo simulation is used to evaluate system performance under stochastic demand conditions [15]. This approach enables assessment of variability and risk beyond deterministic scenarios.

TABLE VII
MONTE CARLO SIMULATION RESULTS

Metric	Value
Mean Stockout Rate (%)	0.02
Mean Service Level (%)	99.98
Mean Final Cash	147159.67
Std Dev of Cash	549.51
Minimum Cash	145992.84
Maximum Cash	148655.11

The computational overhead of Monte Carlo simulation depends on the number of simulation runs and the complexity of the underlying model. In this study, a moderate number of simulations (e.g., 100–200 runs) was used, which provided reliable estimates while maintaining reasonable computational efficiency. For real-time applications, the simulation can be optimized using parallel processing or reduced sampling techniques to ensure scalability and responsiveness.

The results indicate high system reliability, with near-zero stockout rates and consistently high service levels. The relatively low standard deviation of cash suggests stable financial performance, while the range of outcomes reflects inherent uncertainty in demand.

Although advanced models such as Random Forest and LSTM were evaluated, Linear Regression demonstrated more stable and consistent performance for the given dataset. The relatively small dataset size and presence of noise limit the effectiveness of complex models, which tend to overfit or produce unstable predictions. In contrast, Linear Regression provides better generalization and interpretability, making it more suitable for integration into the simulation framework.

Overall, the evaluation confirms that the proposed framework achieves both high operational efficiency and financial stability under uncertain conditions.

Additionally, LSTM models were experimented with; however, they did not outperform simpler regression-based models, likely due to limited dataset size and noise sensitivity [4], [11].

VI. Discussion

The experimental results demonstrate the effectiveness of integrating machine learning with digital twin simulation for supply chain optimization. The proposed framework successfully combines demand forecasting, inventory control, financial modeling, and risk analysis into a unified decision-support system, enabling coordinated decision-making across operational and financial dimensions.

To evaluate the contribution of each component, an ablation study was conducted by incrementally removing elements of the proposed framework. When only forecasting was used without simulation, inventory decisions lacked adaptability, resulting in increased stockouts and inefficient replenishment cycles. When simulation was applied without financial modeling, operational performance improved; however, the absence of financial insights limited the ability to assess cost efficiency and cash flow impact. The full integration of forecasting, simulation, and financial modeling produced the best overall performance, highlighting the importance of joint optimization in supply chain systems.

The forecasting component plays a critical role in enabling data-driven inventory decisions. By capturing overall demand trends while smoothing short-term fluctuations, the model provides stable inputs for downstream simulation. However, this smoothing effect introduces a trade-off between stability and responsiveness, as sudden demand spikes may not be fully captured. Such limitations are closely related to demand amplification issues like the bullwhip effect, where delayed or distorted signals lead to inefficiencies across the supply chain [14].

The superior performance of adaptive inventory policies highlights the importance of dynamic decision-making. Unlike static baseline approaches, adaptive policies continuously adjust reorder points based on predicted demand, resulting in improved service levels and reduced stockout occurrences. This demonstrates that incorporating predictive insights into operational policies significantly enhances system responsiveness, consistent with modern supply chain management practices [6], [9].

From a financial perspective, the integration of cash flow modeling provides a more comprehensive evaluation of supply chain decisions. The results indicate that improved inventory policies not only enhance service levels but also optimize cash utilization and working capital efficiency. However, in certain scenarios, achieving higher service levels requires maintaining slightly increased inventory levels, reflecting a trade-off between liquidity and service reliability. This emphasizes the importance of balancing operational performance with financial constraints in real-world systems [13], [17].

The robustness of the proposed system is further validated through stress testing and Monte Carlo simulation. The ability of the adaptive policy to maintain high service levels across varying scenarios demonstrates resilience against demand uncertainty and external disruptions. At the same time, variability in financial outcomes highlights the inherent trade-offs between risk and performance. While average performance remains stable, the distribution of outcomes suggests that decision-makers must account for uncertainty when selecting inventory strategies [15].

In addition to internal model comparisons, the proposed approach is conceptually benchmarked against traditional statistical methods such as moving averages and classical reorder policies. The results indicate that the ML-driven adaptive policy achieves superior service levels and financial outcomes due to its ability to incorporate predictive insights and simulate dynamic system behavior, which are not captured in conventional approaches.

Furthermore, analysis of forecasting errors indicates that most deviations occur during periods of sudden demand fluctuations. This suggests that while linear models provide stable and interpretable performance, future improvements could involve hybrid or ensemble models to better capture non-linear patterns and extreme variations in demand [11].

Overall, the findings demonstrate that integrating machine learning with digital twin technology provides a scalable and effective solution for modern supply chain management. By enabling predictive, adaptive, and risk-aware decision-making, the proposed framework addresses key limitations of traditional methods and supports more efficient, resilient, and financially optimized operations [1], [2].

VII. Limitations

Despite the promising results, the proposed framework has certain limitations that should be considered. First, the study is conducted on a relatively limited dataset, which may restrict the generalizability of the results to larger and more complex supply chain environments. The performance of machine learning models, particularly deep learning approaches, may improve with access to larger and more diverse datasets.

Second, the current framework assumes simplified supply chain conditions, including fixed lead times and single-echelon inventory management. In real-world scenarios, supply chains often involve multi-echelon structures, dynamic lead times, and additional operational constraints, which are not fully captured in this study.

Third, while Monte Carlo simulation provides valuable insights into demand uncertainty, it introduces additional computational overhead. Although manageable for offline analysis, scalability to real-time decision-making environments may require optimization techniques such as parallel processing or reduced sampling strategies.

Furthermore, the forecasting model primarily focuses on historical demand patterns and does not incorporate external influencing factors such as promotions, weather conditions, or economic indicators. Including such variables could further improve prediction accuracy and system performance.

Finally, the proposed framework is evaluated in a simulated environment, which may not fully reflect all real-world complexities. Practical deployment would require integration with real-time data systems and validation in live operational settings.

VIII. Conclusion

This paper presented an ML-driven digital twin framework for optimizing supply chain operations and cash flow management under demand uncertainty. The proposed system integrates demand forecasting, inventory simulation, financial modeling, and risk analysis into a unified decision-support framework.

The results demonstrate that machine learning-based forecasting effectively captures demand patterns and improves inventory decision-making compared to traditional static approaches. Adaptive inventory policies significantly enhance system performance by reducing stockouts, maintaining stable inventory levels, and optimizing cash flow.

The incorporation of safety stock further improves system reliability by mitigating the impact of demand uncertainty, consistent with classical inventory management principles. Additionally, Monte Carlo simulation provides valuable insights into system variability and risk, enabling more robust decision-making under uncertainty.

Overall, the integration of machine learning with digital twin simulation offers a scalable and practical approach for modern supply chain optimization, aligning with recent advancements in digital twin technologies.

IX. Future Work

Several extensions can be explored to further enhance the proposed framework.

First, advanced machine learning models such as gradient boosting and deep learning architectures can be incorporated to improve forecasting accuracy. Second, multi-product and multi-echelon supply chain modeling can be implemented to better reflect real-world complexity.

Third, reinforcement learning-based policies can be explored to enable dynamic and self-optimizing inventory decisions. Additionally, incorporating real-time data streams and external factors such as promotions, weather, and economic indicators can further improve model responsiveness.

Finally, extending the system to include supplier variability, dynamic lead times, and real-time dashboards can enhance practical applicability and enable deployment in real-world industrial environments.

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